

Processando config 5, seed 4, teste: BlockAlphabit
 xx
 Starting Alphabit: nb = 16777216
 Version: TestU01 1.2.3
 xx

 HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
 Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
 unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
 Sparse = FALSE

Number of bits = n = 16777216
 Number of cells = 2^L = 4
 Expected number per cell = 4194304
 Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	2
Value of the statistic	:	0.39
p-value of test	:	0.82

CPU time used : 00:00:00.03

Generator state:

16777216 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
 Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
 unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
 Sparse = FALSE

Number of bits = n = 16777216
 Number of cells = 2^L = 16
 Expected number per cell = 1048576
 Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 7.71
p-value of test : 0.46

CPU time used : 00:00:00.03

Generator state:

33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 143.40
p-value of test : 0.17

CPU time used : 00:00:00.02

Generator state:

50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32723.56
p-value of test : 0.57

CPU time used : 00:00:00.03

Generator state:

67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 195.08
p-value of test : 0.17

CPU time used : 00:00:00.02

Generator state:

83886080 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 311.32
p-value of test : 0.39

CPU time used : 00:00:00.02

Generator state:

100663296 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.55
p-value of test : 0.29

CPU time used : 00:00:00.03

Generator state:

117440512 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	25.35
p-value of test	:	0.79

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	26.28
p-value of test	:	0.71

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	26.25
p-value of test	:	0.75

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	18.25
p-value of test	:	0.83

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	18.51
p-value of test	:	0.24

CPU time used : 00:00:00.07

Generator state:

134217728 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	60.90
p-value of test	:	0.52

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	71.19
p-value of test	:	0.20

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	164.22
p-value of test	:	0.39

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	46.06
p-value of test	:	0.85

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	23.29
p-value of test	:	0.87

CPU time used : 00:00:00.06

Generator state:
150994944 bits have been read.

===== Summary results of Alphabit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits:	16777216
Number of statistics:	17
Total CPU time:	00:00:00.32

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

```
For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
```

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic            : 1.79
p-value of test                   : 0.41
```

```
-----
CPU time used                     : 00:00:00.03
```

```
Generator state:
```

```
167772160 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
```

Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	8
Value of the statistic	:	5.38
p-value of test	:	0.72

CPU time used : 00:00:00.02

Generator state:

184549376 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	128
Value of the statistic	:	126.50
p-value of test	:	0.52

CPU time used : 00:00:00.00

Generator state:

201326592 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2
```

smultin_MultinomialBitsOver test:

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 0.00097653271, Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom : 32768
Value of the statistic : 32961.33
p-value of test : 0.22
```

```
-----
CPU time used : 00:00:00.03
```

Generator state:

218103808 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2
```

sstring_HammingIndep test:

```
-----
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0
```

Counters with expected numbers >= 10

```
-----
Number of degrees of freedom : 177
Chi-square statistic : 192.81
p-value of test : 0.20
```

```
-----
CPU time used : 00:00:00.02
```

Generator state:

234881024 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305
Chi-square statistic	:	321.68
p-value of test	:	0.24

CPU time used	:	00:00:00.00
---------------	---	-------------

Generator state:

251658240 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic	:	-1.12
p-value of test	:	0.87

CPU time used	:	00:00:00.02
---------------	---	-------------

Generator state:

268435456 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	29.26
p-value of test	:	0.61

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	35.95
p-value of test	:	0.25

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	22.05
p-value of test	:	0.91

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	32.76
p-value of test	:	0.14

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	10.91
p-value of test	:	0.76

CPU time used : 00:00:00.07

Generator state:

285212672 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	63.47
p-value of test	:	0.42

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	69.41
p-value of test	:	0.24

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	149.79
p-value of test	:	0.71

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	74.05
p-value of test	:	0.06

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	36.02
p-value of test	:	0.29

CPU time used : 00:00:00.04

Generator state:

301989888 bits have been read.

===== Summary results of Alhabit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits:	16777216
Number of statistics:	17
Total CPU time:	00:00:00.27

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 4
```

smultin_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic             : 5.28
p-value of test                    : 0.07
```

```
-----
CPU time used                      : 00:00:00.02
```

Generator state:

318767104 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 4
```

smultin_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
```

Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

$\Delta = 1$, $\mu = 2.0861626e-07$, $\sigma = 1$

Test Results for $\Delta = 1.0000$

Number of degrees of freedom	:	8
Value of the statistic	:	4.59
p-value of test	:	0.80

CPU time used : 00:00:00.03

Generator state:
335544320 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

 $N = 1$, $n = 16777216$, $r = 0$, $s = 32$, $L = 8$,
Sparse = FALSE

Number of bits = $n = 16777216$
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

$\Delta = 1$, $\mu = 3.784895e-06$, $\sigma = 1$

Test Results for $\Delta = 1.0000$

Number of degrees of freedom	:	128
Value of the statistic	:	123.95
p-value of test	:	0.58

CPU time used : 00:00:00.02

Generator state:
352321536 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32441.02
p-value of test : 0.90

CPU time used : 00:00:00.03

Generator state:

369098752 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 160.48
p-value of test : 0.81

CPU time used : 00:00:00.02

Generator state:

385875968 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305
Chi-square statistic	:	330.26
p-value of test	:	0.15

CPU time used : 00:00:00.00

Generator state:

402653184 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic	:	-1.77
p-value of test	:	0.96

CPU time used : 00:00:00.02

Generator state:

419430400 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	53.25
p-value of test	:	0.01

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	45.11
p-value of test	:	0.05

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	33.73
p-value of test	:	0.38

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	28.63
p-value of test	:	0.28

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	15.18
p-value of test	:	0.44

CPU time used : 00:00:00.07

Generator state:

436207616 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	54.93
p-value of test	:	0.73

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	62.47
p-value of test	:	0.46

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	170.54
p-value of test	:	0.27

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	55.70
p-value of test	:	0.52

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	34.92
p-value of test	:	0.33

CPU time used : 00:00:00.04

Generator state:

452984576 bits have been read.

===== Summary results of Alphabit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits:	16777216

Number of statistics: 17
Total CPU time: 00:00:00.27

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alhabit: nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

smultin_MultinomialBitsOver test:

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.9802323e-08, Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom : 2
Value of the statistic : 2.72
p-value of test : 0.26
```

```
-----
CPU time used : 00:00:00.02
```

Generator state:
469761792 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

smultin_MultinomialBitsOver test:

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.0861626e-07, Sigma = 1
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom : 8
Value of the statistic       : 3.96
p-value of test              : 0.86
```

CPU time used : 00:00:00.02

Generator state:

486539008 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

```
Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 3.784895e-06, Sigma = 1
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom : 128
Value of the statistic       : 143.21
p-value of test              : 0.17
```

CPU time used : 00:00:00.00

Generator state:

503316224 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32940.49
p-value of test : 0.25

CPU time used : 00:00:00.03

Generator state:

520093440 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 174.07
p-value of test : 0.55

CPU time used : 00:00:00.00

Generator state:
536870656 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305
Chi-square statistic	:	358.78
p-value of test	:	0.02

CPU time used : 00:00:00.02

Generator state:
553647872 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -1.70
p-value of test : 0.96

CPU time used : 00:00:00.00

Generator state:
570425088 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	42.75
p-value of test	:	0.10

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	27.70
p-value of test	:	0.64

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	31.04
p-value of test	:	0.52

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	19.78
p-value of test	:	0.76

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	13.94
p-value of test	:	0.53

CPU time used : 00:00:00.07

Generator state:

587202304 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	47.79
p-value of test	:	0.91

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	53.12
p-value of test	:	0.78

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	152.28
p-value of test	:	0.66

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	75.81
p-value of test	:	0.05

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	19.32
p-value of test	:	0.96

CPU time used : 00:00:00.04

Generator state:
603979264 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3

Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25

All tests were passed

XXX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XXX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 3.28
p-value of test : 0.19

CPU time used : 00:00:00.03

Generator state:
620756480 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 9.97
p-value of test : 0.27

CPU time used : 00:00:00.00

Generator state:
637533696 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 153.14
p-value of test : 0.06

CPU time used : 00:00:00.02

Generator state:
654310912 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	: 32768
Value of the statistic	: 32946.00
p-value of test	: 0.24

CPU time used : 00:00:00.02

Generator state:

671088128 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	: 177
Chi-square statistic	: 173.48
p-value of test	: 0.56

CPU time used : 00:00:00.02

Generator state:
687865344 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305
Chi-square statistic	:	304.42
p-value of test	:	0.50

CPU time used : 00:00:00.00

Generator state:
704642560 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic	:	-0.19
p-value of test	:	0.58

CPU time used : 00:00:00.03

Generator state:
721419776 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	39.44
p-value of test	:	0.17

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	24.51
p-value of test	:	0.79

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	15.06
p-value of test	:	0.9952

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	34.46
p-value of test	:	0.10

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	12.93
p-value of test	:	0.61

CPU time used : 00:00:00.06

Generator state:
738196992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	53.14
p-value of test	:	0.78

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	55.03
p-value of test	:	0.72

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	148.49
p-value of test	:	0.73

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	64.34
p-value of test	:	0.24

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	35.63
p-value of test	:	0.30

CPU time used : 00:00:00.06

Generator state:
754973952 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25

All tests were passed

XXX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XXX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 3.65
p-value of test : 0.16

CPU time used : 00:00:00.00

Generator state:

771751168 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 9.59
p-value of test : 0.29

CPU time used : 00:00:00.02

Generator state:

788528384 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 137.05
p-value of test : 0.28

CPU time used : 00:00:00.02

Generator state:
805305600 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 33045.99
p-value of test : 0.14

CPU time used : 00:00:00.02

Generator state:
822082816 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 150.01
p-value of test : 0.93

CPU time used : 00:00:00.02

Generator state:
838860032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 305.00
p-value of test : 0.49

CPU time used : 00:00:00.02

Generator state:
855637248 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 2.01
p-value of test : 0.02

CPU time used : 00:00:00.00

Generator state:
872414464 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	41.31
p-value of test	:	0.13

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	36.49
p-value of test	:	0.23

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	23.41
p-value of test	:	0.86

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	20.12
p-value of test	:	0.74

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	18.64
p-value of test	:	0.23

CPU time used : 00:00:00.07

Generator state:

889191680 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config5_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	53.58
p-value of test	:	0.77

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	76.41
p-value of test	:	0.10

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	140.22
p-value of test	:	0.87

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	61.01
p-value of test	:	0.33

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	33.28
p-value of test	:	0.40

CPU time used : 00:00:00.04

Generator state:

905968640 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25

All tests were passed